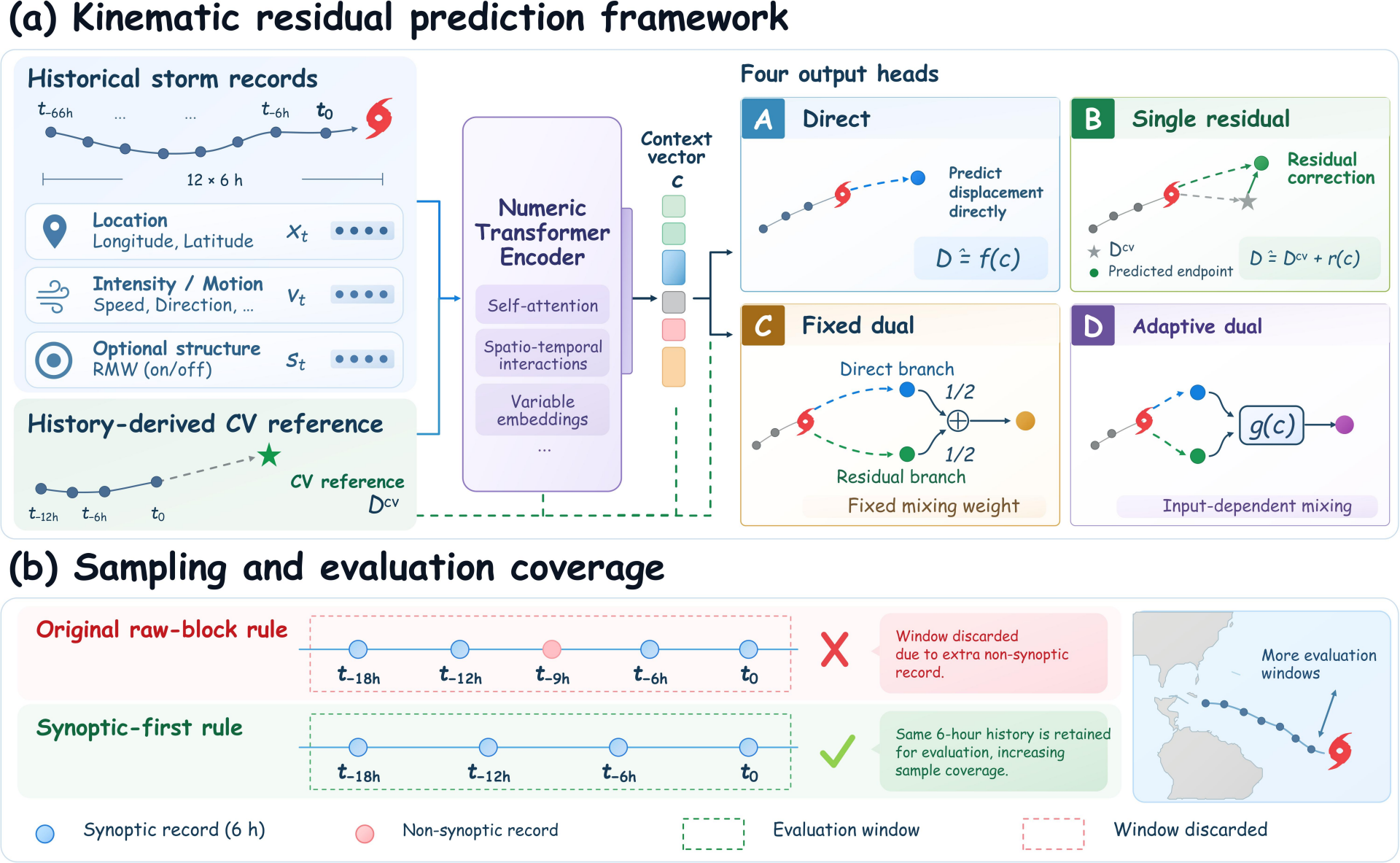

(a) Kinematic residual prediction framework
Historical storm records
t-66h
t-6h
t0
12 × 6 h
Location
Longitude, Latitude
Xt
Intensity / Motion
Speed, Direction, ...
Vt
Optional structure
RMW (on/off)
St
History-derived CV reference
CV reference
Dcv
t-12h
t-6h
t0
Numeric Transformer Encoder
Self-attention
Spatio-temporal interactions
Variable embeddings
...
Context vector
c
Four output heads
A Direct
Predict displacement directly
D ≈ f(c)
B Single residual
Residual correction
Dcv
Predicted endpoint
D ≈ Dcv + r(c)
C Fixed dual
Direct branch
1/2
1/2
Residual branch
Fixed mixing weight
D Adaptive dual
g(c)
Input-dependent mixing
(b) Sampling and evaluation coverage
Original raw-block rule
t-18h
t-12h
t-9h
t-6h
t0
Window discarded due to extra non-synoptic record.
Synoptic-first rule
t-18h
t-12h
t-6h
t0
Same 6-hour history is retained for evaluation, increasing sample coverage.
More evaluation windows
Synoptic record (6 h)
Non-synoptic record
Evaluation window
Window discarded